SWE 270P Assignment 2
Professor Iftekhar Ahmed

Instructions for Submission
This is a Group Assignment. Please make sure to assign yourself to your group in the People tab. Only one submission per group is required.

Submit your **Python code (.ipynb)** to Canvas for Assignment 2 before the deadline. As part of the assignment, you must schedule a time slot with the TA to demo your working application. **No late submissions will be accepted**.

1. Python Code

- **File name format**: `SWE270P-A2.ipynb`
- Must be in `.ipynb` format
- Each Python file must include a comment at the top with your name and student ID
- Make sure to write comments explaining each function/method implemented

2. Demo

- Schedule a time slot with the TA and run a full demo of your application.
- The demo should clearly show the process of:
- Uploading the code image
- Transcribing the code snippet
- Running static analysis to explain the code and suggest improvements

Plagiarism warning: Please be aware that although you may help each other understand the concepts that we are practicing, you may not provide each other with or receive the answers to these questions. Submitting someone else's work will cause me to take punitive action, including reporting such activity to the academic dishonesty office.

Discussion forum: There is an Ed discussion thread for this homework. Please post clarifying questions there so that everyone can benefit from the answers and so that the TAs and I do not need to answer the same questions repeatedly. I also encourage you to participate there: ask questions, and offer help to each other. Again, I am saying "help", not "give the answer". I will be monitoring the board and will recognize and reward participation in the Participation portion of the course grade.



Assignment 2: Code Whiteboard Tutor

Main goal: Your main goal is to use a multimodal Large Language Model (LLM) to build a UI application that allows users to upload a photo of their handwritten Python code and receive suggestions for code improvements. The main functionalities that must be included are:

- Transcribing handwritten Python code into a code snippet in
- Running static analysis and explaining bugs (if any) in natural language
- Suggesting bug fixes, improvements, or efficiency tweaks to the code snippet

Resources: We recommend using Gradio, which allows you to directly display the UI of the app you are building. For the multimodal LLM, we have included *"gemini-3.5-flash-lite"* in the demo code. You are welcome to use any other open-source multimodal LLM that is compatible with your computing resources. However, we strongly discourage you from paying for model usage.

Note: You will **not** be graded on how well the multimodal LLM performs. Instead, grading will be based on the completeness of your assignment and the demo.

Instructions for Assignment 2:

Step 1. Download and unzip the **SWE270P_A2.zip,** and read the instructions in `SWE270P_A2demo.ipynb`.The zip folder contains examples of handwritten code and the corresponding test cases. You can use the Google Colab environment or your preferred IDE.

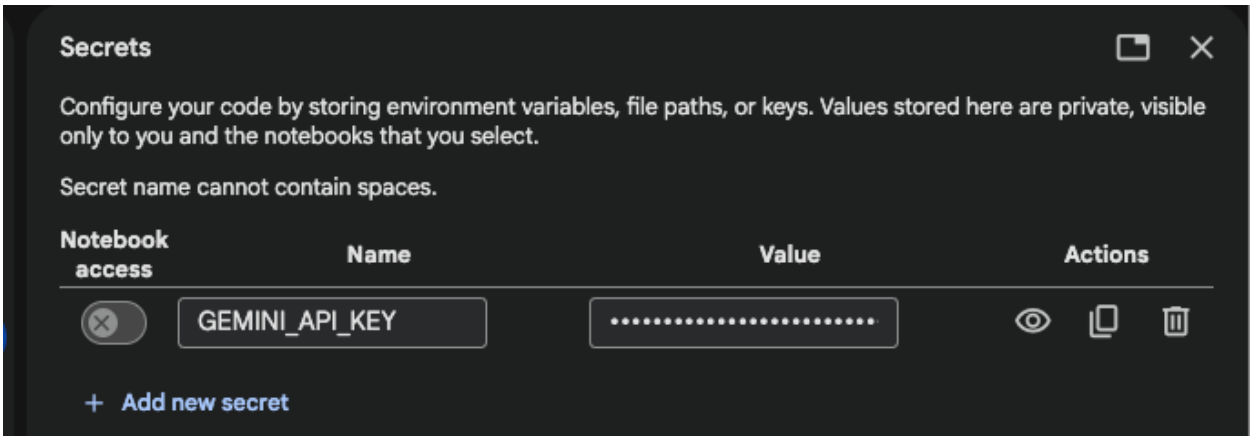
Step 2. Create your Google Gemini API key (to use the same model as in the demo code).

Step 3. Add your API key to **your copy**. For Google Colab notebook, use the Secrets tab (located on the left navigation bar, see picture below for reference) to add your API key.

Step 4. Write your code and test it using the images provided in the *code_images_example* folder within the ZIP file. You can also write the same function in your own handwriting to test it.

Step 5. Schedule a time slot with the TA to demo your working Gradio application. You must use the version of the code you submitted. Be prepared to run your application for multiple rounds using images provided by the TA.

Make sure to test that your Gradio application works under different scenarios (e.g., various handwriting styles, colored pens), run additional test cases to ensure the robustness of your code, and be ready to answer questions during the demo to demonstrate your understanding of the code and the overall functionality of your application.



Helpful Guidelines to Complete Your Assignment

(Extra Credit) Extra points will be awarded if your implementation supports multiple programming languages.

The grading criteria are available on the course Canvas page. You are welcome to use the template code we provide or create your own. In both cases, you must include comments in the code to explain the purpose and key implementation details of each function or method.

The assignment will be graded based on the code (including the written comments) and the demo (a working UI demonstration). Make sure to follow the specified submission format. Failure to do so may result in a score of 0 due to incompatibility. No late submissions will be accepted for this assignment.